H-B-181b
October 2, 1963

SUPERSEDING
Fed. Spec. H-B-160
April 13, 1948 and
Fed. Spec. H-B-181a
November 10, 1947

FEDERAL SPECIFICATION

BRUSH, CHASSIS AND RUNNING GEAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers oiled palmetto fiber, chassis and running-gear cleaning brushes.

1.2 Classification.

1.2.1 *Type and size.* Brushes covered by this specification shall be of one type and one size.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

PPP-B-636—Box, Fiberboard.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-W-6110—Wood; Determination of Moisture Content of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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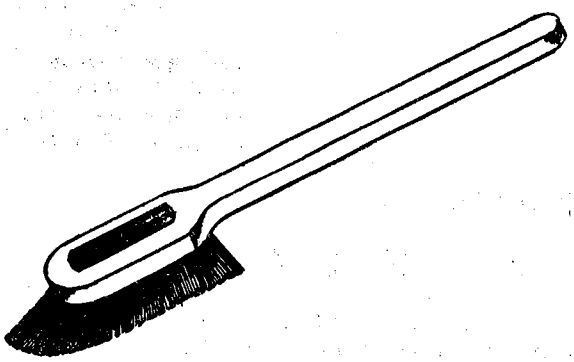


FIGURE 1.—Brush, Chassis and Running Gear

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials. Materials shall conform to the applicable specifications and requirements specified hereinafter.

3.2.1 *Oiled palmetto fiber.* The oiled palmetto fiber shall show no perceptible change in stiffness when tested as specified in 4.4.3.

3.2.2 *Wood.* Wood shall be close-grained and straight-grained beech, birch, maple, or sycamore. The wood shall be thoroughly and uniformly dried to a moisture content not exceeding 12 percent at time of assembly. Moisture content shall be determined in accordance with 4.4.4.

3.2.3 *Staple wire.* Staple wire shall be not less than 0.047-inch diameter, commercial-steel staple wire, treated to resist corrosion in accordance with the manufacturer's standard practice.

3.3 Construction. The brushes shall conform to the requirements specified herein and in table I.

TABLE I.—*Physical requirements*

Block:	
Length, over-all, minimum, inches	19-1/2
Length, brush part, minimum, inches	3-1/8
Width, brush part, minimum, inches	2-1/2
Width, handle at end, minimum, inch	3/4
Thickness, minimum, inches	1
Holes:	
Number, minimum	56
Diameter, minimum, inch	9/32
Fiber:	
Weight, minimum, ounces	3-7/8
Length, clear of block, minimum, inches	2
Slant of fiber at end and sides, approx.	30 degrees

3.3.1 *Block.* The block shall be made from wood specified in 3.2.2. The top edges of the entire block and the bottom edges of the handle part shall be rounded to a radius of approximately 1/8 inch. The bottom edges of the brush part of the block shall be rounded to a radius of approximately 1/2 inch. The brush end shall be rounded on a radius of approximately 3-1/2 inches. The end of the handle shall be rounded on a radius of approximately 1/2 inch. A hole not less than 3/16 inch in diameter shall be located on the center line of the handle approximately 1 inch from the end of the handle.

3.3.2 *Staple set.* The fiber specified in 3.2.1 shall be cut to double length and center-folded in quantities to insure individual tufts of such size to fit snugly when inserted in the holes provided in the block. Stapling shall be accomplished so that the staples, made from wire specified in 3.2.3, and fiber are driven to the bottom of the holes, with the staple securely imbedded in the wood.

3.3.3 *Retention of fiber.* The fiber shall be so secured in the block that when tested as

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specified in 4.4.1, it shall not be possible to remove the fiber from the block without breaking the fiber.

3.4 Weight of fiber. The weight of the fiber shall be not less than specified in table I when tested in accordance with 4.4.2.

3.5 Finish. Wood blocks shall have a plain sanded or tumbled smooth finish or shall be coated with varnish, lacquer or other suitable clear organic material in accordance with the manufacturer's standard commercial practice.

3.6 Marking for identification. The supplier shall permanently and legibly indent his name, or trademark of such known character that it is easily identifiable with said manufacturer, on the handle of each brush.

3.7 Workmanship. The finished brushes shall conform to the quality and grade of product established by this specification. Component parts shall be neatly finished and assembled. There shall be no exposed sharp wire, sharp edges, or splinters.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. The preproduction sample shall be examined in accordance with 4.3.3.1, dimensions specified, and shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 *Component and material inspection.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.3.2 *Intermediate testing.* The wood blocks shall be tested as specified in 4.4.4 to determine compliance with 3.2.2. The sample unit shall be one block and the lot size shall be expressed in units of blocks offered for acceptance at one time. The test shall be made immediately prior to assembly and finish. The requirements shall be applicable to the lot average and the results shall be reported numerically to the nearest 0.1 percent. There shall be one determination made per unit and the inspection level shall be level L3.

4.3.3 *End-item inspection.* The sample unit for this inspection shall be one completely fabricated brush. All of the brushes offered for acceptance at one time shall be considered a lot for the purpose of inspection.

4.3.3.1 *Visual examination.* Examination of the end item shall be in accordance with the defects list set forth as follows. The inspection level shall be level I and the acceptable quality level (AQL), shall be 4.0, expressed in defects per hundred units.

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Examine	Defect
Construction and workmanship:	
Block	Edges not rounded as specified, end of handle not rounded, hole missing.
	Not finished as specified.
	Wood component not smoothly finished; splintered, torn or loose grain.
Stapling	Staples not driven into the bottom of the holes.
Fiber	Fiber not evenly distributed in brush head.
	Fiber not clean; evidence of dirt or foreign matter.
	Brush head not evenly trimmed.
Marking	Identification marking omitted, incomplete, incorrect.

4.3.3.2 *Dimensional examination.* Inspection shall be made of the end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be L5 and the AQL shall be 6.5, expressed in defects per hundred units.

4.3.3.3 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table II herein, from each lot presented for examination. The sample unit shall be one brush and the lot size shall be expressed in units of brushes. The inspection level and AQL shall be as indicated in table II.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored in accordance with defects list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers offered for delivery at one time. The inspection level shall be L4 and the AQL shall be 6.5, expressed in defects per hundred units.

Examination	Defect
Markings (exterior and interior container)	Missing; incorrect, illegible, of improper size, location, sequence or method of application.
Material	Component missing, damaged or otherwise defective.
Packaging	Contents more or less than specified.
	Box not compact or well filled.
	Closure omitted or poorly performed.
Packing	Contents more or less than specified.
	Closure omitted, not as specified, or poorly accomplished.

4.4 Test.

4.4.1 *Fiber retention.* Individual strands of fiber shall be held between the thumb and forefinger and pulled to determine compliance with 3.3.3.

4.4.2 *Fiber weight.* The fiber shall be removed from the brush and conditioned for not less than 16 hours in air at a relative humidity of 50 percent, plus or minus 4 per-

TABLE II.— *Instructions for testing*

Characteristic	Specification reference		Rqmts. Appl. to		Number determinations per sample unit	Results reported as		Inspection level	AQL
	Requirement	Test method	sample unit	Lot average		Pass or fail	Numerically to nearest		
Retention of fiber	3.3.3	4.4.1	X	—	10	X	—	L3	4.0
Weight of fiber	3.4 and Table 1	4.4.2	—	X	1	—	0.1 ounce	L3	—
Stiffness of fiber	3.2.1	4.4.3	X	—	10	X	—	L3	4.0

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cent, and at a temperature of 73°F., plus or minus 2°F. The conditioned fiber shall then be weighed in the same atmosphere to determine compliance with 3.4.

4.4.3 *Stiffness test.* The fiber, tested as specified in 4.4.2, shall be soaked in boiling water (212°F.) for not less than one hour. The fiber shall be removed from the water and placed face up in a holding device so that it is held firmly in a manner and position similar to a finished brush. A finished brush manufactured in accordance with this specification shall be attached in a face-up position alongside the fiber. The fiber in the holding device and the brush shall be depressed with the palm of the hand not less than 10 times respectively for comparison to determine compliance with 3.2.1.

4.4.4 *Moisture content.* The moisture content of wood specified in 3.2.2 shall be determined in accordance with either method of MIL-W-6110.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Twelve brushes shall be packaged in a snug-fitting fiberboard box conforming to type I or II, class 1, style RSC of PPP-B-636, except that the minimum bursting strength of the fiberboard shall be 200 pounds per square inch (p.s.i.). Brushes shall be packaged six wide and two deep, each brush reversed (head to handle) and a sheet of rigid paperboard or fiberboard inserted between the two layers of brushes. Each box shall be closed by covering all seams full length with strips of 2-1/2-inch-minimum-width tape conforming to class 2 of UU-T-111 or PPP-T-76. The center strip shall extend a minimum of 2-1/2 inches onto the end panels. Alternatively, the bottom flaps may be sealed with adhesive or by metal stitching.

5.1.2 *Level C.* Brushes shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Sixty brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.* Twelve brushes, packaged as specified in 5.1.1, shall require no overpacking. Sixty brushes, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and ship-

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ping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended for use in the cleaning of vehicular chassis and running gear.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) When preproduction sample is required (3.1).

(*c*) Applicable levels of packaging and packing (5.1 and 5.2).

6.3 Illustration. The illustration shown herein as figure 1 is for the purpose of identification and is not intended to preclude any brush which otherwise meets the requirements of this specification.

6.4 Level A packaging and level B packing should be used by ordering agencies only when shipments are known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Brushes, vegetable or wood fibre. Carload minimum weight 30,000 pounds.

Motor:

Brushes, vegetable or wood fibre. Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—GL

Navy—Y

Air Force—MOA

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